



INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH MOHALI
(Established by Ministry of Human Resource Development, Govt. of India)
Sector 81, S A S Nagar, Manauli PO Punjab 140306

MID-SEMESTER EXAM

EES 202 Date: 11.09.2024

Introduction to Atmospheric and Climate Sciences

Total Marks: 20 (2 pages)

Please write your name and roll number on this sheet before you commence answering.

Choose correct option for Section A and answer questions in Sections B and C in the space provided after each question after reading the question carefully.

Name of Student:

Roll Number:

Section A: Choose the correct option ($1 \times 5 = 5$ Marks; 1 for correct answer and -1 for wrong answer)

1) What is the weight of 1 mol of dry air in the Earth's present-day atmosphere?

(a) 28.0 g/mol (b) 28.96 g/mol (c) 32.2 g/mol (d) 44.0 g/mole

2) What is the total mass of earth's atmosphere? Consider $P=1000$ mbar; $R_E=6400$ km; $g=10\text{ms}^{-2}$

(a) 5.15×10^{18} kg (b) 5.25×10^{18} g (c) 5.2×10^{18} kg (d) 54×10^{22} g

3) What is the number density of air at sea level where $P=1000$ hPa and $T=300$ K

(a) 2.4×10^{25} molecules m^{-3} (b) 2.4×10^{26} molecules cm^{-3} (c) 2.7×10^{26} molecules m^{-3} (d) 10^{10} molecules L^{-1}

4) The mixing ratio of hydrogen in the atmosphere is:

(a) 50 ppt (b) 100 ppm (c) 10 ppb (d) 500 ppb

5) The value of the scale height (H) of Earth's atmosphere taking $T=250$ K & $g=10\text{ms}^{-2}$ is:

(a) 7.4 m (b) 7.4 km (c) 10 km (d) 10 m

Section B: II. Very short answer type questions ($2 \times 5 = 10$ marks)

Q6) Neon has a fixed mixing ratio in the atmosphere. How would you expect its number density measured in surface air to vary between day and night? How would you expect its partial pressure measured in surface air to vary between day and night? ($1+1=2$ Marks)

Q7) Explain with examples the contextual relevance of reporting the ambient abundance of chemical atmospheric constituents in terms of i) mixing ratio ii) number density iii) mass concentrations iv) partial pressure. ($4 \times 0.5=2$ Marks)

Q8) Derive the relationship between mass concentration and mixing ratio of an atmospheric gas "X" possessing a molecular weight "M". (2 Marks)



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Q9) Calculate the amount of oxygen in TgO in the Earth's atmosphere, given that mixing ratio of oxygen is 0.21. (Take mean sea-level pressure = 1000 hPa ; $g = 10 \text{ m s}^{-2}$; Radius of Earth = 6400 km; Take mass of 1 mole of air = 29 g mol^{-1}) (2 Marks)

Q10) If water had a lower specific heat capacity than land, then how would you expect the land seas breeze effect to be during mid-afternoon and mid-night in terms of directionality (2 Marks)

Section C:

Q11) Derive the barometric law for the Earth's atmosphere and sketch the vertical profile of pressure in the Earth's atmosphere as a fraction of the surface pressure P_0 . (3+2= 5 Marks)